Spurious Predictability from an Embedding Index Error: A Case Study in Financial Analog Forecasting

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Abstract

Adaptive, multi-month model development can generate statistically significant forecasting results even when the underlying effect is an artifact of the pipeline itself rather than of the data. We document one such case in delay-embedding-based financial forecasting. Testing four structurally different novelty and anomaly detectors across seven asset classes, we initially found forecast accuracy substantially higher on days flagged as locally atypical, a result that survived walk-forward cross-validation, permutation testing, and a genuine out-of-sample holdout. The result turned out to be an artifact of a one-step index error: the embedding's most recent component was numerically identical to the first step of the future return the model was trying to predict. We confirm this diagnosis at several levels of pipeline complexity, from a closed-form algebraic proof to a pipeline-free correlation test, a null control on synthetic white noise, and a power test showing the corrected pipeline still detects genuine designed-in signal. All four detectors were affected identically, which points to the shared embedding rather than to any one detection method. We propose a two-part diagnostic protocol, a null-noise control paired with surrogate-data testing, that would have caught this error in hours rather than months, and we recommend it as standard practice before reporting a positive result in this class of study.

Keywords: spurious predictability, data leakage, delay embedding, analog forecasting, surrogate data, anomaly detection

1. Introduction

Short-horizon financial return forecasting via historical-analog methods, delay-embedding recent returns into a low-dimensional state space, locating

the k nearest historical analogs, and averaging their realized continuations, draws on a long-standing technique with roots in the Analog Ensemble literature from atmospheric and space-weather forecasting [5] and in the broader nonlinear-dynamics tradition of state-space reconstruction [17, 12]. A natural extension of this idea, pursued across a large and active recent literature, is to condition forecast confidence or forecast usage on an auxiliary novelty or anomaly-detection score computed on the same embedded state. The intuition is simple: states the model has rarely seen before, measured by some notion of atypicality in the embedded space, might be more or less predictable than familiar ones.

This paper began as an attempt to formalize and test that intuition properly. Using a diffusion-model reconstruction score, and, for robustness, three classical novelty detectors (Isolation Forest, Local Outlier Factor, and Mahalanobis distance in PCA space) as the atypicality signal, we found across seven asset classes, an equity index, two energy benchmarks, two precious metals, and two cryptocurrencies, that forecast accuracy conditional on high atypicality was substantially and significantly higher than conditional on low atypicality. The result held up under nested walk-forward evaluation, combinatorial purged cross-validation, formal permutation testing [8], and a genuine single-shot 20% holdout that had never been touched during model development. It even motivated a dynamical-systems explanation involving local Lyapunov exponents, which itself replicated on a second holdout.

The result was wrong. Section 4 describes how we found out and, more importantly, why none of the safeguards above caught it. We think the diagnostic protocol that eventually exposed the error, and that we recommend other researchers run before reporting a similar finding, is this paper's most useful contribution.

This kind of failure is not unique to our pipeline or to financial applications of machine learning specifically. Harvey et al. [10] documented the scale of factor proliferation in the asset-pricing literature and the role data mining plays in it, and Hou et al. [11] showed that a large share of published anomalies fail to replicate under more conservative testing. What distinguishes the case reported here is that the spurious effect did not survive sloppy validation; it survived careful validation, including a form of out-of-sample testing that is often treated as close to definitive. That is the part we think is worth documenting in detail.

1.1. Contributions

We document a previously unreported class of index error in delay-embedding-based forecasting pipelines, under which the embedding's most recent feature is numerically identical to part of the forecast target, and we show that it survives methods normally considered strong safeguards against overfitting: walk-forward cross-validation, permutation testing, and a genuine out-of-sample holdout. We verify this diagnosis at six increasing levels of pipeline complexity, from a closed-form algebraic proof to a null-hypothesis control on synthetic data with a provably known absence of structure. To be precise about what this buys us: these are six checks at different levels of abstraction, not six independent sources of evidence, since checks 1 and 3 below are two representations of the same underlying fact. We show that the error affected all four structurally different detectors identically, which establishes that the apparent finding was a property of the shared embedding convention rather than of any one detection method. Finally, we propose a lightweight, two-part diagnostic protocol, a pure-noise null control together with AAFT surrogate-data testing, that would have caught this error immediately, and that we recommend as a standard pre-submission check for this class of method.

2. Related Work

Delay embedding and analog forecasting. The analog-forecasting technique underlying this investigation, locating the k nearest historical states to the current embedded state and pooling their verifying continuations, is structurally identical to the Analog Ensemble (AnEn) framework developed in the atmospheric-forecasting literature. Hamill and Whitaker [9] introduced the underlying analog-correction technique for precipitation reforecasts; Delle Monache et al. [5] extended it with a Kalman-filter/analog post-processing scheme, and Delle Monache et al. [6] formalized the probabilistic AnEn method as a way of generating an ensemble forecast directly from a single deterministic trajectory. Alessandrini et al. [1] demonstrate AnEn's practical value for short-term wind-power forecasting, illustrating the framework's reach beyond its original precipitation application. The state-space reconstruction underlying the embedding itself follows the standard delay-coordinate approach of Takens [17] and the broader nonlinear time-series methodology surveyed in Kantz and Schreiber [12]. To our knowledge, the broader project this paper is drawn from is the first to apply the AnEn

framework, under that name, to financial return series rather than to a meteorological or physical system.

Spurious predictability and leakage in financial machine learning. The integration of machine learning into empirical finance has expanded the set of models and features available to researchers, but standard validation methods were largely developed for approximately i.i.d. data and can fail under the temporal dependence and causal-ordering constraints that financial time series impose [13]. The closest precedent to this paper's diagnostic contribution is Nikolopoulos [14], who proposes a falsification framework for financial ML pipelines that tests complete modeling workflows against synthetic reference distributions with known predictability, a goal closely related to the null-noise control used here, though developed independently and applied to a different class of method. On the statistical side, the multiplicity concerns raised by testing many model and detector variants on the same panel are formalized by White [20]'s Reality Check for data snooping and by the garden of forking paths framework of Gelman and Loken [7], applied directly to financial-anomaly research by Coqueret [4].

Surrogate-data testing and the chaos-in-markets debate. The second component of our diagnostic protocol draws directly on the surrogate-data methodology introduced by Theiler et al. [19] for testing nonlinearity against a matched-linear null, later refined by Schreiber and Schmitz [15, 16] with the Iterative Amplitude-Adjusted Fourier Transform (IAAFT), a closer-to-exact version of the AAFT surrogates used here. This methodology underlies a decades-long, still-unsettled literature on whether financial return series exhibit genuine deterministic chaos or linear stochastic structure that merely resembles it, critically surveyed in Barnett and Serletis [3]. Studies applying surrogate-data and related nonlinearity tests to individual equity indices have reached conflicting conclusions depending on denoising and test choice, which underscores that this remains an open methodological question rather than a settled one. The lesson we take from that literature, that apparent nonlinear structure in a detector statistic has to be checked against a matched linear surrogate null before it can be attributed to genuine dynamics, is precisely the lesson our own investigation initially failed to apply to its own core predictability claim, and not only to its exploratory dynamical-systems explanation.

3. Data and Baseline Method

3.1. Data

Daily close prices for seven assets, downloaded from Yahoo Finance: Brent crude (`BZ=F`), the S&P 500 index (`^GSPC`), gold (`GC=F`), silver (`SI=F`), WTI crude (`CL=F`), Bitcoin (`BTC-USD`), and Ethereum (`ETH-USD`), using the maximum available history per asset (8–20 years depending on listing date).

3.2. Delay embedding and analog selection

Log returns $r_t = \log(P_t/P_{t-1})$ are delay-embedded into $\mathbb{R}^M$ with $M = 3$, $\tau = 1$: $x_j = (r_t, r_{t-\tau}, r_{t-2\tau})$. For a forecast origin, the $K = 7$ nearest historical analogs, by a learned feature distance described in Section 3.3, are selected causally, excluding a Theiler window of 10 days around the origin to avoid trivial temporal auto-matches [18]. The forecast is the sign of the mean $H = 3$-day forward return realized by the selected analogs.

3.3. Atypicality detectors

Four detectors score each origin's atypicality: the reconstruction loss (ELBO) of a small denoising diffusion model, a 2-layer MLP with hidden width 32, trained causally on the embedded history; Isolation Forest; Local Outlier Factor; and Mahalanobis distance in a PCA-reduced space with Ledoit-Wolf shrinkage. All four are fit causally and retrained periodically on history up to the current origin only.

3.4. Evaluation

We use nested nonoverlapping walk-forward folds with step size H, a 2000-resample permutation test for the accuracy-gap statistic defined by a median split on the atypicality score, Fisher's method to combine p-values across the seven-asset panel, and, separately, a single untouched 20% holdout per asset, evaluated exactly once.

4. The Error: Discovery, Diagnosis, and Fix

4.1. The original result

Using the method of Section 3, we found a substantial, statistically significant gap in forecast accuracy between high- and low-atypicality origins. The diffusion-ELBO detector, for example, showed a +8.3 percentage-point accuracy gap on Brent crude ($p < 0.0001$), and the three classical detectors

showed comparable or larger gaps on the same and other assets (Table 1 reports the full pre-fix comparison across all four detectors and seven assets). A genuine single-shot holdout test, touched for the first time only after model development was frozen, reproduced the effect at the panel level, with a Fisher-combined $p = 3.8 \times 10^{-5}$ across the seven assets. We additionally proposed and validated a dynamical-systems explanation involving local Lyapunov exponents, which itself replicated on a second, independent holdout and proved robust to parameter choices and to splitting the sample by calendar era.

4.2. The surrogate-data control that exposed the problem

The problem surfaced during what we intended as a routine robustness check of the dynamical-systems explanation, not of the core claim. We generated AAFT surrogate return series, matching the real data's amplitude distribution and linear power spectrum exactly while randomizing phases to destroy any nonlinear temporal structure, and re-ran the local-Lyapunov analysis on them. The purely linear surrogates reproduced, and in fact exceeded, the real data's effect size, for both a cheap k-NN atypicality proxy and the production diffusion-ELBO detector. Taken on its own, that result does not require positing a coding error; the surrogate-data literature is full of cases where apparent nonlinear structure turns out to be linear. It did, however, prompt us to apply the same control to the paper's actual central claim, the atypicality-conditioned accuracy gap, rather than only to the auxiliary Lyapunov statistic.

4.3. The decisive control: pure white noise

We generated literal i.i.d. Gaussian white noise, a process with, by construction, no genuine forecastable structure of any kind, and ran the identical pipeline on it. The pipeline reported a significant atypicality-conditioned accuracy gap of 5–7 percentage points, with permutation p-values as low as the resolution floor of a 2000-resample test, on every one of three initial noise draws; this was extended to ten trials across two assets in Section 5, with the same result each time. A process with zero genuine predictive structure cannot produce a genuine accuracy edge of that size by definition, so the only remaining explanation is a defect in the measurement procedure itself.

4.4. Root cause

In plain terms, the pipeline's "most recent known return," the feature used to judge how unusual today looks and which historical days resemble it, was not actually known yet at the moment the forecast was made. It was one day into the future the model was supposedly predicting.

More precisely, with origin index j, embedding length M, and delay τ, the project's convention located "today's price" at array index $p = j + (M-1)\tau$ and defined the forecast target as the H-step return beginning at p. The embedding's most recent component, $x_j^{(0)}$, is constructed from $\texttt{log_ret}[j + (M-1)\tau]$, which is exactly $\texttt{log_ret}[p]$: the first of the H returns summed to produce the forecast target itself. The correct convention is $p = j + M\tau$, which places the embedding entirely in the past relative to the forecast window, with zero gap and zero overlap.

4.5. Why none of the standard safeguards caught it

The error sits in every fold's feature-construction code identically, in both the training and the held-out portions of the data, because it is a fixed, deterministic property of the index arithmetic rather than something learned from or overfit to any particular sample. That is precisely why each of the three safeguards this literature relies on most heavily missed it. Walk-forward cross-validation protects against overfitting to one historical window, but offers no protection against a structural bias present identically in every window. A genuine single-shot holdout protects against re-using information learned during model selection, but offers no protection against a bias baked into the feature-construction code itself, which is applied, by construction, the same way to the holdout period as to everything else. Permutation testing protects against mistaking sampling noise for a real effect, but offers no protection when the effect is a deterministic artifact that reappears, with the same mechanism intact, in every permutation.

Each of these three tools is, individually, considered strong evidence in the forecasting literature, and none of them is designed to catch a fixed arithmetic bias in feature construction, because none of them varies the feature-construction code itself. A fourth tool does: comparing the pipeline's output on real data against its output on data engineered to contain no real structure at all (Section 4.3, formalized as a general protocol in Section 7.2) tests the feature-construction code directly, rather than the sample it happens to be run on.

5. Verification

Given what the diagnosis implies, the central claim of a multi-month investigation turning out to be an artifact, we did not accept it on the strength of any single check. The six checks below span increasing levels of pipeline complexity, from a closed-form algebraic identity to the full production pipeline. Checks 1 and 3 are two representations of the same underlying fact rather than independent evidence, and are listed separately only because they operate at different levels of abstraction. Full detail and numbers are available in Findings 51–52 of the accompanying code repository.

1. **Exact algebraic identity.** We hand-derived and numerically confirmed that $x_j^{(0)}$ equals `log_ret`$[p]$ exactly, not approximately, on a tiny synthetic price array.
2. **Generalization across the real series.** The same overlap held over 33 origins (47 in a fully independent re-implementation) spread across real Brent history, universally under the old formula and never under the corrected one.
3. **A pipeline-free, model-free direct test.** The sign of $x_j^{(0)}$ matched the sign of the actual future return in 68.3% of folds under the old convention, against 50.1%, chance, under the fix. That 68.3% rather than roughly 100% is itself consistent with the diagnosis, since the leaked component is only one of three summands making up the target.
4. **A null control repeated ten times** across two assets and two random seeds. Under the fixed pipeline, the mean permutation p-value across trials was 0.496, close to the theoretical expectation of exactly 0.5 under a true null, with a false-positive rate matching the nominal 5% level.
5. **A fully independent, from-scratch script** with no imports from the project's own codebase, using a fresh data download and hand-written embedding and index arithmetic, reproduced the identical result. This rules out a shared misreading of the code baked into the investigation's own tooling.
6. **A power test with a known, designed-in signal.** A synthetic series with an explicit, by-construction magnitude-conditional mean-reversion rule, after any return exceeding one standard deviation, the next three returns are deliberately biased in the opposite direction, was correctly detected by the fixed pipeline (accuracy rose from 52.2% to

58.2%, gap $p = 0.011$), confirming that the fix did not simply desensitize the method.

We also ran a systematic six-point leakage audit beyond the specific bug described above, covering data-feature temporal ordering, embedding-normalization causality, train/test shuffling, target-feature alignment, residual embedding overlap, and detector self-inclusion. One further issue turned up, smaller and independent of the main one: the embedding's scale normalization is computed over the full series rather than causally. We disclose it here and show it does not materially affect the result (Brent: +2.4pp moves to −0.1pp under a corrected causal-scale variant, both non-significant; S&P 500: −1.2pp stays at −1.2pp).

6. Results After Correction

Table 1: Diffusion-ELBO accuracy gap, corrected pipeline, by asset.

Asset	Accuracy gap	p-value
Brent	+2.4pp	0.176
S&P 500	−1.2pp	0.692
Gold	−5.4pp	0.985
Bitcoin	−1.7pp	0.720
Ethereum	+1.2pp	0.352

No accuracy gap reaches significance, and several point estimates are negative. Extending the comparison to all four detectors across the full seven-asset panel, 28 tests in total, finds exactly 1 of 28 cells below $p = 0.05$ (Ethereum/Mahalanobis-PCA, $p = 0.025$), which is indistinguishable from the roughly 1.4 false positives expected by chance alone at $\alpha = 0.05$ across 28 independent tests (the full table is reported as Finding 53 in the accompanying repository).

The genuine single-shot holdout test, re-run with the corrected pipeline, tells the same story at the panel level. Every asset is individually non-significant, and Fisher's combined p-value across the seven-asset panel is **0.288**, against the original, uncorrected pipeline's $p = 3.8 \times 10^{-5}$ on the identical protocol. The strongest-looking evidence in the original investigation collapses, once the leak is removed, to a result indistinguishable from chance.

7. Discussion

7.1. Why this is worth publishing

We think a null result reached through careful, multiply-verified diagnosis of a subtle and previously undocumented methodological error is more useful to the forecasting community than the original, incorrect, positive finding would have been. The error is structural to a widely used class of method, delay embedding paired with auxiliary novelty or anomaly scoring, and is not specific to our particular implementation choices, so it is likely to recur wherever the same convention is used. We also show, concretely, that the standard toolkit for guarding against overfitting in this literature, walk-forward validation, permutation testing, and even genuine out-of-sample holdout testing, did not detect it. That is itself an actionable finding about the limits of those safeguards for this specific failure mode, and it echoes a broader point made by Bailey et al. [2] about the gap between formal rigor and actual protection against overfitting in backtest-driven research. Finally, we offer a lightweight, generally applicable diagnostic, a null-noise control paired with surrogate-data testing, that would have caught this class of error cheaply, before a researcher spends months building an elaborate positive-finding narrative on top of it, as we did.

7.2. A recommended diagnostic protocol

We recommend that any study pairing delay embedding with an auxiliary detection score for financial, or other, time-series forecasting run two checks before reporting a positive result. The first is a null-data control: apply the identical pipeline to synthetic data with provably zero real structure, i.i.d. noise matching the real series' scale is sufficient, and confirm that the reported effect does not appear. The second is a surrogate-data control: apply the identical pipeline to AAFT, or comparable, surrogates of the real series, to separate genuine structure from an artifact that matched linear noise can reproduce on its own. Both checks are inexpensive relative to the modeling effort typically invested in this class of study, and together they would have caught the error documented here within hours rather than months.

7.3. Limitations

This paper documents one specific error in one specific implementation of a standard method family. We are not claiming the error is present in any other published study; we are claiming that the general class of convention it

belongs to, using the embedding's most-recent-return component without an explicit, verified alignment check against the forecast target's own starting index, carries this risk, and that the diagnostic protocol we propose is a cheap, general-purpose safeguard whether or not any specific paper happens to contain this exact error.

8. Conclusion

We set out to validate the claim that market states flagged as locally atypical by an unsupervised novelty detector are more forecastable than typical states. The claim passed every standard validation step used in this literature. It did not survive a null-data control, because the apparent effect was, in fact, a one-step index error that gave the forecasting model partial access to its own target. We confirm this diagnosis at six increasing levels of pipeline complexity and show that the corrected pipeline retains full power to detect genuine designed-in signal, which rules out the possibility that the correction simply broke the method. We offer the diagnostic protocol that exposed the error, a null-noise control paired with a surrogate-data test, both standard tools in adjacent literatures but, to our knowledge, not yet standard practice in this specific subfield, as this paper's main contribution.

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